

Lecture Unit 7.5

Parallel overheads and Amdahl's law

Performance degradation

SAB: section 1.4.5

- A number of factors can contribute to real-world parallel performance being less than the idealised peak performance:
 - **S**tarvation: insufficient parallel work to keep the processors busy (insufficient parallel work, uneven distribution of work)
 - **L**atency: time taken for information to travel from one part of the system to another (e.g. accessing memory, sending messages)
 - **O**verhead: work required in addition to the computation (e.g. starting and stopping OpenMP parallel regions)
 - **W**aiting: multiple threads/processes accessing a shared resource (e.g. contention for memory or network bandwidth)

Parallel overheads

- Parallel overheads are the extra work required to run a program in parallel (in addition to the computation)
- MPI has explicit overheads (e.g. extra lines of code to distribute work, send messages etc)
- There are overheads in OpenMP although they are less obvious e.g.
 - Overhead associated with starting and ending parallel regions
 - Loop scheduling overhead

Amdahl's law with overheads

- We can't eliminate these overheads but we can seek to minimise them and understand their impact
- We can include overheads in a revised version of Amdahl's law
- For a simple model we will add a parallel overhead which is incurred n_p times (e.g. each time a parallel region is started)
- Let the parallel overhead be v which is incurred n_p times, hence the total overhead is

$$V = n_p v$$

Amdahl's law with overheads

- We will add an overhead $n_p v$ to the parallel run time
- The serial run time is still given by:

$$T_0 = sT_0 + pT_0$$

- The parallel run time is now given by:

$$T_N = sT_0 + \frac{pT_0}{N} + n_p v$$

T_0 is a time e.g. in second
 v is also a time e.g. in seconds

- The parallel speed-up (S_N) is given by:

$$S_N = \frac{T_0}{T_N} = \frac{sT_0 + pT_0}{sT_0 + \frac{pT_0}{N} + n_p v} = \frac{s + p}{s + \frac{p}{N} + \frac{n_p v}{T_0}} = \frac{1}{s + \frac{p}{N} + \frac{n_p v}{T_0}}$$

Amdahl's law with overheads

$$S_N = \frac{1}{s + \frac{p}{N} + \frac{n_p v}{T_0}}$$

- This expression is Amdahl's law with an extra term in the denominator
- $\frac{n_p v}{T_0}$ is the total overhead as a fraction of the serial run time
- When $\frac{n_p v}{T_0}$ is significant compared to s the overhead affects the speed-up

$$\frac{n_p v}{T_0} = \frac{V}{T_0}$$

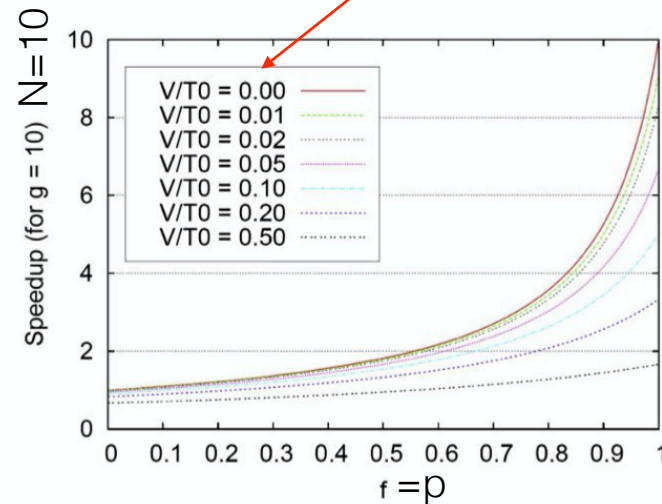


FIGURE 6.5 The speedup for various overhead ratios as a function of the fraction of code that can be accelerated for a fixed

SAB figure 6.5

Unit Summary

- To run a program in parallel extra work required in addition to the computation (parallel overhead)
- Parallel overheads cannot be eliminated but need to be minimised to improve performance and scaling
- We can include an overhead term in Amdahl's law to model the effect of parallel overheads on scaling